

TrES-1b

R-2044

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_120402 · created 2026-08-08T23:27:27+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2456019.92426 +/- 0.00079 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1618 +/- 0.0075
Transit depth (Rp/Rs)^2	2.62 +/- 0.24 [%]
Orbital Inclination (inc)	88.39 +/- 0.72
Airmass coefficient 1 (a1)	1.341 +/- 0.011
Airmass coefficient 2 (a2)	-0.1552 +/- 0.0038
Scatter in the residuals of the lightcurve fit is	0.83 %
Best Comparison Star	None
Optimal Aperture	4.49
Optimal Annulus	13.97
Transit Duration (day)	0.1026 +/- 0.0035

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1618 $\pm$ 0.0075 vs 0.1358 (deviation 3.47 σ)
Duration fit vs published	0.1026 d vs 0.1042 d (deviation 0.45 σ)
Mid-time O-C	2.1 min
Depth SNR	21.6
Residual scatter	0.83 %

Light curve

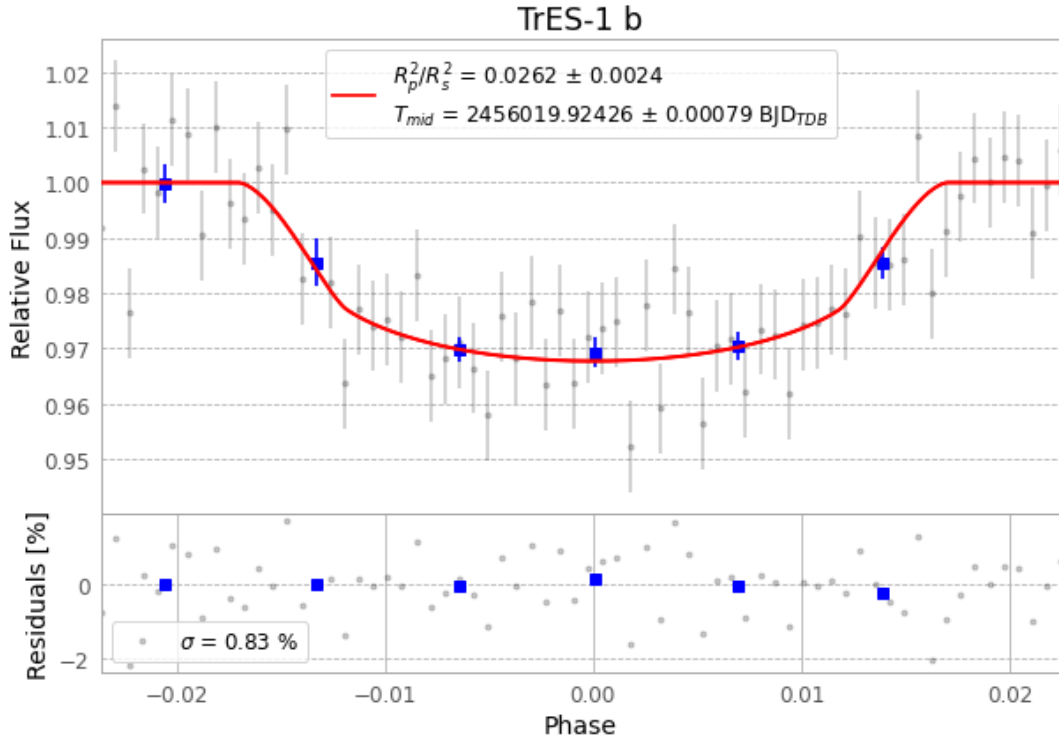


Figure 1 — FinalLightCurve_TrES-1 b_2012-04-02.png (SHA-256 bf1f19d2b33a9f54...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [282, 149], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 fead2b890fd0b560...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

68 FITS frames (45 MB), TRES-1120402082712.FITS → TRES-1120402114812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-120402.log (2e2707e570dd...), FinalLightCurve_TrES-1 b_2012-04-02.png (bf1f19d2b33a...), FinalLightCurve_TrES-1 b_2012-04-02.csv (beeb004521ed...)

Compute resources

Wall time 263.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-08 23:27Z.